Tangent lines, SOS and the uvw method

Kevin Hu
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L5-X3 Level 5 Advanced · Algebra · Beyond the textbook

Three methods strong solvers use on symmetric inequalities, namely the tangent line trick with a case split, SOS with the criteria that decide its sign, Schur and SOS-Schur, and the uvw reduction to two equal variables or one zero.
By the end you can apply the tangent line trick on the whole allowed range, splitting into cases where the line fails; write a symmetric inequality in SOS or SOS-Schur form and finish the proof with the standard criteria or Schur's inequality; state and prove the uvw lemma and use the lemma to reduce a symmetric inequality to two-variable checks.
Prerequisites: Classical inequalities (L4-AL2). **Contests:** BMO Round 2, EGMO and IMO selection, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Let $a, b, c \geq 0$ with $a + b + c = 3$. Prove that $4(ab + bc + ca) \leq 9 + 3abc$.

Equality holds at $(1, 1, 1)$ and also at $(\frac{3}{2}, \frac{3}{2}, 0)$. The second equality case rules out most of L4-AL2. A chain of AM-GM steps, or a sum of tangent line bounds, is sharp only where every step is sharp, and the steps we know there are sharp only at $a = b = c$. The inequality is Schur's inequality in disguise, and is the simplest example of a symmetric inequality that needs one of the techniques of this unit, namely the tangent line trick in its full form, SOS (sum of squares) together with the criteria that decide its sign, Schur's inequality in the forms that combine with SOS, and the uvw method, which reduces a symmetric inequality in three variables to the cases where two variables are equal or one is zero.

None of these is in a school textbook and none is deep. They are the working methods of strong inequality solvers. The tangent line trick is folklore of olympiad training. The SOS criteria were collected by Vietnamese problem solvers, and Pham Kim Hung's *Secrets in Inequalities* (2007) is the standard reference. The uvw method takes its name from Tejs Knudsen's article *The uvw method*, written for AoPS, and was popularised there by Michael Rozenberg. The method is close to Vasile Cîrtoaje's equal variable method. We assume L4-AL2 throughout. For three variables write $p = a + b + c$, $q = ab + bc + ca$ and $r = abc$, and write $\sum$ for a sum over the three cyclic shifts of (a, b, c).

1.1 The tangent line trick, in full

Theorem 1.
Let f be defined on an interval I, let $s \in I$, and suppose $f(x) \leq f(s) + m(x - s)$ for all $x \in I$. If $x_1, \ldots, x_n \in I$ and $x_1 + \cdots + x_n = ns$, then $f(x_1) + \cdots + f(x_n) \leq nf(s)$. If s is an interior point of I and f is differentiable at s, the only slope that can work is $m = f'(s)$. (The same

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holds with every inequality reversed.)
Proof. Add the n inequalities. The terms $m(x_i - s)$ add up to 0. For the slope, $g(x) = f(s) + m(x - s) - f(x)$ is non-negative on I with $g(s) = 0$, so g has a minimum at the interior point s, and $g'(s) = 0$.

Three remarks turn this into a method. First, when f is a rational function and $m = f'(s)$, the difference $f(s) + m(x - s) - f(x)$ has a factor $(x - s)^2$. Divide this factor out and study the sign of what is left on the whole of I. Second, if the sign is right only on part of I, split into the case where every x_i lies in the good part and the case where some x_i does not. In the second case a crude bound is usually enough, because the bad region is far from the equality case. Third, the constraint need not be a sum. If the constraint is $\sum g(x_i) = n\,g(s)$, look for $f(x) \le f(s) + m\big(g(x) - g(s)\big)$ instead.

Theorem 1 works for any number of variables, so this is the first thing to try with four or more. Its limitation is that the theorem proves only inequalities whose sole equality case is $x_1 = \cdots = x_n = s$.

Example 1.
Let $a, b, c \ge 0$ with $a + b + c = 3$. Prove that

$$\frac{a^2+1}{a^3+1} + \frac{b^2+1}{b^3+1} + \frac{c^2+1}{c^3+1} \le 3.$$

Solution.
Let $f(x) = \frac{x^2+1}{x^3+1}$. Then $f(1) = 1$ and the quotient rule gives $f'(x) = -\frac{x(x^3+3x-2)}{(x^3+1)^2}$, so $f'(1) = -\frac{1}{2}$ and the candidate line is $y = \frac{3-x}{2}$. Since $(x+1)(x^2 - x + 1) = x^3 + 1$,

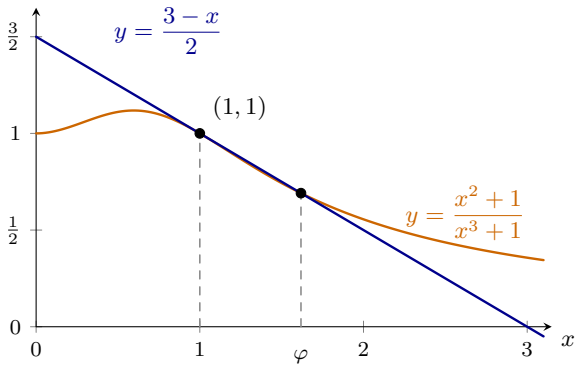
$$\frac{3-x}{2} - f(x) = \frac{(x-1)^2(1 + x - x^2)}{2(x^3+1)}.$$

The factor $1 + x - x^2$ is positive for $0 \le x < \varphi = \frac{1+\sqrt{5}}{2}$ and negative for $x > \varphi$, so the line lies above the curve only on $[0, \varphi]$, as the figure shows. The tangent line trick alone fails, and we split.
If $a, b, c \le \frac{3}{2}$, then, as $\frac{3}{2} < \varphi$, adding the three tangent line bounds gives a total of at most $\frac{9-(a+b+c)}{2} = 3$.
Otherwise one variable, say a, exceeds $\frac{3}{2}$. For $x \ge 1$ we have $x^3 + 3x - 2 > 0$, so f is decreasing on $[1, \infty)$ and $f(a) < f(\frac{3}{2}) = \frac{26}{35}$. For the other two variables we use $f(x) \le \frac{9}{8}$ for all $x \ge 0$, which is $9x^3 - 8x^2 + 1 \ge 0$. This is clear for $x \ge \frac{8}{9}$. For $0 \le x < \frac{8}{9}$, AM-GM on $\frac{9x}{2}, \frac{9x}{2}, 8 - 9x$ gives

$$8x^2 - 9x^3 = \tfrac{4}{81}\left(\tfrac{9x}{2}\right)^2 (8 - 9x) \le \tfrac{4}{81}\left(\tfrac{8}{3}\right)^3 = \tfrac{2048}{2187} < 1.$$

Hence the sum is less than $\frac{26}{35} + \frac{9}{4} = \frac{419}{140} < 3$. Equality holds only at $a = b = c = 1$.
The split point $\frac{3}{2}$ was chosen because $\frac{3}{2}$ lies below φ and keeps $f(a)$ small enough. The margin $\frac{419}{140}$ against 3 is small. Had the check failed, we would have moved the split point.

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1.2 SOS and the criteria

A symmetric inequality in three variables with equality at $a = b = c$ can usually be written, after subtracting one side from the other, as

$$S = S_a(b-c)^2 + S_b(c-a)^2 + S_c(a-b)^2 \geq 0,$$

where S_a, S_b, S_c are expressions in a, b, c. The standard pieces are

$$a^2+b^2+c^2-q = \sum \tfrac{1}{2}(a-b)^2, \qquad \sum \frac{a}{b+c} - \frac{3}{2} = \sum \frac{(a-b)^2}{2(a+c)(b+c)}, \qquad p\left(\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c}\right) - 9 = \sum \frac{(a-b)^2}{ab},$$

each checked by a short computation. To find the SOS form, write each part of the inequality as a combination of these and collect the coefficient of $(a-b)^2$ as S_c. If all three coefficients are non-negative we are done. Usually one of them is not, and then the ordering of the variables helps.

Theorem 2 (SOS criteria).
Let $S = S_a(b-c)^2 + S_b(c-a)^2 + S_c(a-b)^2$.

1. If $S_a, S_b, S_c \geq 0$, then $S \geq 0$.
2. If $a \geq b \geq c$ and $S_b \geq 0$, $S_a + S_b \geq 0$, $S_c + S_b \geq 0$, then $S \geq 0$.
3. If $a \geq b \geq c \geq 0$, $b > 0$, and $S_b \geq 0$, $S_c \geq 0$, $a^2 S_b + b^2 S_a \geq 0$, then $S \geq 0$.

Proof. (1) is trivial. For (2), $a - c = (a-b) + (b-c)$ with both terms non-negative, so $(a-c)^2 \geq (a-b)^2 + (b-c)^2$, and as $S_b \geq 0$,

$$S \geq (S_a + S_b)(b-c)^2 + (S_c + S_b)(a-b)^2 \geq 0.$$

For (3), if $S_a \geq 0$ use (1), and if $b = c$ then $S = S_b(a-c)^2 + S_c(a-b)^2 \geq 0$. Otherwise $b(a-c) - a(b-c) = c(a-b) \geq 0$ gives $\frac{a-c}{b-c} \geq \frac{a}{b}$, so

$$S \geq S_a(b-c)^2 + S_b(a-c)^2 \geq (b-c)^2\left(S_a + \frac{a^2}{b^2}S_b\right) = \frac{(b-c)^2}{b^2}\left(a^2 S_b + b^2 S_a\right) \geq 0.$$

In (2) and (3) you may assume $a \geq b \geq c$ only when S is symmetric, that is, when S_a, S_b, S_c come from one formula by permuting the variables. For a cyclic expression you may assume only that a is the largest, and both orders of b, c must be checked.

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Example 2.
Let a, b, c be non-negative reals, at most one of them zero. Prove that

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} \geq 1+\frac{a^2+b^2+c^2}{2(ab+bc+ca)}.$$

Solution.
Both sides equal $\frac{3}{2}$ at $a=b=c$, and both equal 2 at $(1,1,0)$. The standard route is to clear denominators. This gives a symmetric inequality of degree 5 which is also sharp at $(1,1,0)$, so AM-GM steps that are sharp only at $a=b=c$ cannot finish the proof. After expanding, we are left with a symmetric quintic whose proof needs Schur-type ideas anyway. SOS is much shorter. Subtracting $\frac{3}{2}$ from both sides and using the standard pieces, the inequality becomes

$$\sum \frac{(a-b)^2}{2(a+c)(b+c)} \geq \sum \frac{(a-b)^2}{4q}, \qquad \text{that is} \qquad \sum S_c(a-b)^2 \geq 0 \text{ with } S_c=\frac{q-c^2}{4q(a+c)(b+c)}.$$

Criterion (1) fails. At $(4,1,1)$, $q-a^2=-7$. The expression is symmetric, so we may assume $a \geq b \geq c$. Then $q-c^2 \geq ab-c^2 \geq 0$ and $q-b^2=b(a-b)+c(a+b) \geq 0$, so $S_b, S_c \geq 0$. Finally

$$S_a+S_b=\frac{(q-a^2)(b+c)+(q-b^2)(a+c)}{4q(a+b)(b+c)(c+a)}=\frac{2c(2ab+bc+ca)}{4q(a+b)(b+c)(c+a)} \geq 0,$$

because $q(a+b+2c)=a^2b+ab^2+a^2c+b^2c+2bc^2+2c^2a+4abc$. Criterion (2) proves the inequality.
For equality, the proof of (2) shows $S \geq (S_a+S_b)(b-c)^2+(S_b+S_c)(a-b)^2$. If $c>0$ both brackets are positive, so $a=b=c$. If $c=0$, then $S_a+S_b=0$ but $S_b+S_c \geq S_c=\frac{1}{4ab}>0$, so $a=b$. Equality holds exactly at $a=b=c$ and at $a=b, c=0$ (and permutations).

1.3 Schur's inequality and SOS-Schur

Theorem 3 (Schur).
For $a, b, c \geq 0$ and $t>0$,

$$a^t(a-b)(a-c)+b^t(b-a)(b-c)+c^t(c-a)(c-b) \geq 0,$$

with equality if and only if $a=b=c$ or two of the variables are equal and the third is 0.
Proof. By symmetry let $a \geq b \geq c$. The first two terms add up to $(a-b)\left[a^t(a-c)-b^t(b-c)\right] \geq 0$, since $a^t \geq b^t$ and $a-c \geq b-c \geq 0$, and the third term is $c^t(a-c)(b-c) \geq 0$. The equality cases follow by asking when both parts vanish.

For $t=1$ the following forms are all the same expression, and each is checked by expanding:

$$\sum a(a-b)(a-c)=p^3-4pq+9r=\sum \tfrac{1}{2}(b+c-a)(b-c)^2=(a+b-c)(a-b)^2+c(a-c)(b-c).$$

The opening problem is the second form with $p=3$. For $t=2$ the pqr form is $p^4-5p^2q+4q^2+6pr \geq 0$. The third form is the SOS form, and criterion (2) proves this form with $S_a=\frac{1}{2}(b+c-a)$.

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The last form is the SOS-Schur form. If $c = \min(a,b,c)$, then $(a-c)(b-c) \geq 0$, so any expression written as $M(a-b)^2 + N(a-c)(b-c)$ with $M, N \geq 0$ is non-negative. The basic instance is $a^2+b^2+c^2-q = (a-b)^2 + (a-c)(b-c)$. The same idea with general weights is the following generalisation, usually named after Vornicu.

Theorem 4 (Vornicu-Schur).
Let $a \geq b \geq c$ and $x, y, z \geq 0$, and suppose $x \geq y$ or $z \geq y$. Then $x(a-b)(a-c) + y(b-a)(b-c) + z(c-a)(c-b) \geq 0$.
Proof. If $x \geq y$, the first two terms add up to $(a-b)\big[x(a-c) - y(b-c)\big] \geq y(a-b)\big[(a-c) - (b-c)\big] = y(a-b)^2 \geq 0$, and the third is $z(a-c)(b-c) \geq 0$. If $z \geq y$, the last two terms add up to $(b-c)\big[z(a-c) - y(a-b)\big] \geq y(b-c)^2 \geq 0$, and the first is $x(a-b)(a-c) \geq 0$.

Example 3.
Non-negative reals a, b, c satisfy $ab + bc + ca = 1$. Prove that $(a+b+c)(a+b+c+3abc) \geq 4$.

Solution.
Equality holds at $a = b = c = \frac{1}{\sqrt{3}}$ and at $(1,1,0)$, so we expect Schur. The condition is not a sum, so we homogenise with $q = 1$. The claim is $p^2 + 3pr \geq 4$, and multiplying p^2 by q and 4 by q^2 gives the degree 4 inequality

$$F = p^2 q + 3pr - 4q^2 \geq 0.$$

Now look for weights. A symmetric polynomial of degree 4 has the form $Ap^4 + Bp^2q + Cq^2 + Dpr$, so two such polynomials are equal as soon as they agree at the four triples $(1,0,0)$, $(1,1,0)$, $(2,1,0)$, $(1,1,1)$ (the four resulting linear equations in A, B, C, D have a unique solution). Comparing at these triples shows

$$F = \sum a(b+c)(a-b)(a-c).$$

Let $a \geq b \geq c$. The weights $x = a(b+c) = q - bc$, $y = q - ca$, $z = q - ab$ satisfy $x \geq y \geq z \geq 0$, so Theorem 4 gives $F \geq 0$.
For the equality cases the SOS-Schur form is more convenient. With $c = \min(a,b,c)$ one checks in the same way that

$$F = (q - c^2)(a-b)^2 + c(a+b)(a-c)(b-c),$$

and both coefficients are non-negative. If $c > 0$, the second term forces $c = a$ or $c = b$, and then the first (with $q - c^2 > 0$) forces $a = b$. If $c = 0$, then $F = ab(a-b)^2$ and $ab = q = 1$ force $a = b = 1$. So equality holds exactly at $a = b = c = \frac{1}{\sqrt{3}}$ and at the permutations of $(1,1,0)$.

1.4 The uvw method

The name comes from writing $p = 3u$, $q = 3v^2$, $r = w^3$. We keep p, q, r. A symmetric expression in a, b, c is a function of p, q, r, and the method rests on one fact about the cubic whose roots are a, b, c.

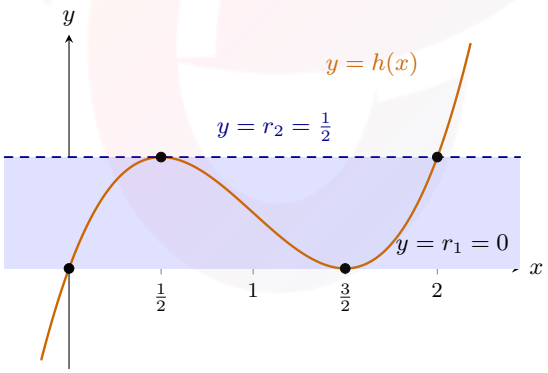
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Theorem 5 (uvw).
Let p, q be real with $p^2 > 3q$, and let $h(x) = x^3 - px^2 + qx$.

1. The values of $r = abc$ over all real triples with $a + b + c = p$ and $ab + bc + ca = q$ form a closed interval $[r_1, r_2]$, and a triple with $r = r_1$ or $r = r_2$ has two equal entries.
2. If also $p, q \geq 0$, the values of r over non-negative such triples form the interval $[\max(r_1, 0), r_2]$, and a triple at the lower end has two equal entries or an entry equal to 0.

(If $p^2 = 3q$ the only real triple is $a = b = c$, and if $p^2 < 3q$ there is none.)
Proof. The numbers a, b, c are the roots of $x^3 - px^2 + qx - r$, that is, the solutions of $h(x) = r$, counted with multiplicity. Since $h'(x) = 3x^2 - 2px + q$ has discriminant $4(p^2 - 3q) > 0$, h' has roots $x_1 < x_2$, and h increases on $(-\infty, x_1]$, decreases on $[x_1, x_2]$ and increases on $[x_2, \infty)$. If $h(x_2) < r < h(x_1)$, the intermediate value theorem gives a root in each of the three intervals. If $r = h(x_1)$, then $h(x) - r$ and its derivative vanish at x_1, so x_1 is a double root, and the third root lies beyond x_2. Similarly for $r = h(x_2)$. If $r > h(x_1)$ or $r < h(x_2)$, the equation has exactly one real root, which is simple, so the other two roots are not real. Hence (1) holds with $r_1 = h(x_2)$, $r_2 = h(x_1)$. For (2), if $p, q, r \geq 0$ and $x < 0$, every term of $x^3 - px^2 + qx - r$ is at most 0 and $x^3 < 0$, so all roots are non-negative. Conversely non-negative roots give $r \geq 0$.

The figure shows h for $p = 3$, $q = \frac{9}{4}$. The line $y = \frac{1}{2}$ touches the graph at $x = \frac{1}{2}$ and meets the graph again at $x = 2$, giving the triple $(\frac{1}{2}, \frac{1}{2}, 2)$ at the top of the range. At the bottom, $r_1 = 0$ comes from $(\frac{3}{2}, \frac{3}{2}, 0)$.



Consequently, if for each fixed p, q an expression $\Phi(p, q, r)$ is monotone in r, or concave in r, then its minimum over triples with those p, q is attained at an end of the interval, that is, when two variables are equal, or, for non-negative variables, when one is zero. A symmetric polynomial of degree d is a combination of terms $p^i q^j r^k$ with $i + 2j + 3k = d$, so for $d \leq 5$ the polynomial is linear in r (the degree argument). Hence a symmetric polynomial inequality of degree at most 5 in non-negative variables needs to be checked only at triples (t, t, s) and $(t, s, 0)$. For degrees 6 to 8 the expression is quadratic in r, and you need the sign of the r^2 coefficient, or a monotonicity argument on the actual range of r.

Example 4.
Non-negative reals a, b, c satisfy $a + b + c = 3$. Find the minimum value of $(1 + a^2)(1 + b^2)(1 + c^2)$.

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Solution.
The standard first guess is the centre, which gives 8, and the standard way to prove this would be a tangent line bound $\ln(1+x^2) \geq \ln 2 + (x-1)$ summed over the three variables. That bound fails at $x = 2$, since $\ln 5 < \ln 2 + 1$, and the failure is real. At $(\frac{1}{2}, \frac{1}{2}, 2)$ the product is $\frac{25}{16} \cdot 5 = \frac{125}{16} < 8$. So the minimum is not at the centre, and no argument sharp only at $a = b = c$ can find the minimum.
With uvw, expanding gives $(1+a^2)(1+b^2)(1+c^2) = 1 + \sum a^2 + \sum a^2b^2 + r^2$, and $\sum a^2 = p^2 - 2q$, $\sum a^2b^2 = q^2 - 2pr$, so with $p = 3$ the product is

$$\Phi = r^2 - 6r + q^2 - 2q + 10.$$

For fixed q this is a quadratic in r which decreases for $r < 3$, and AM-GM gives $r \leq (\frac{p}{3})^3 = 1$. So for fixed q the minimum is at the largest r, which by Theorem 5 comes from a triple with two equal entries. Put $(a, b, c) = (t, t, 3-2t)$ with $0 \leq t \leq \frac{3}{2}$ and $g(t) = (1+t^2)^2(10 - 12t + 4t^2)$. Then

$$g'(t) = 12(1+t^2)(t-1)^2(2t-1),$$

so g decreases on $[0, \frac{1}{2}]$ and increases on $[\frac{1}{2}, \frac{3}{2}]$, and $g(t) \geq g(\frac{1}{2}) = \frac{125}{16}$. The minimum is $\frac{125}{16}$, attained at the permutations of $(\frac{1}{2}, \frac{1}{2}, 2)$.
The degree here is 6, so the degree argument does not apply directly. The sign of the r^2 coefficient is wrong for a minimum, and the step that rescues the method is the bound $r \leq 1$, which places the whole allowed interval on the decreasing side of the parabola. This is the slice $q = \frac{9}{4}$ of the figure.

1.5 What a marker needs to see

1. For the tangent line, state the one-variable inequality, its factorised form, and the range on which you use the inequality, with the case split if the factor changes sign.
2. For SOS, write out the identity with S_a, S_b, S_c, enough intermediate lines (or the four-point check of Example 3) for the marker to verify the identity, and the reason the ordering $a \geq b \geq c$ is allowed.
3. Schur's inequality may be quoted by name with t stated. Theorem 4 and any SOS-Schur form should be proved. Each takes three lines.
4. uvw is not a standard result at BMO2 or the IMO, so do not quote the method as a black box. Expect a marker to accept the method only when the statement is precise and the proof of Theorem 5 is written out, which takes a paragraph. Say exactly what is fixed, why the expression is monotone or concave in r, and what the range of r is. Then do the two-variable checks in full, since that is where the marks are.
5. For a "find the minimum" question, give a proof of the bound and a triple that attains the bound.

Common traps.

- Checking a tangent line bound near the equality point but not on the whole range (Example 1).

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- Assuming $a \geq b \geq c$ for an expression that is only cyclic.
- Forgetting the zero case in uvw. For non-negative variables the lower end of the range of r may be 0 rather than a triple with two equal entries.
- Treating a degree 6 expression as if the expression were linear in r (Example 4 shows what to do instead).
- Applying uvw when the constraint itself involves r, as in $a^2+b^2+c^2+abc=4$, without first deciding which two of p, q, r to fix.
- With positive (rather than non-negative) variables, the boundary case $c = 0$ is not attained, so a bound there is an infimum, not a minimum.

1.6 How to spot this in a contest

1. A sum $\sum f(x_i)$ with $\sum x_i$ fixed, equality at the centre, possibly four or more variables. Use the tangent line, and if the factor changes sign, split the range.
2. Equality at $(1,1,1)$ and also at $(1,1,0)$. Use Schur, SOS with criteria (2) or (3), or SOS-Schur.
3. Fractions with denominators such as $b+c$ or ab. Use SOS with the standard pieces.
4. Symmetric in three variables, polynomial of degree at most 5 (or any function of p, q, r monotone in r), with p or q fixed. Use uvw, then check (t,t,s) and $(t,s,0)$.
5. A "find the minimum" where the centre may not be the answer. Evaluate along $(t,t,3-2t)$ before trying to prove anything.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Let a, b, c, d be real numbers (not necessarily positive) with $a+b+c+d=4$. Prove that

$$\frac{1}{a^2-a+5}+\frac{1}{b^2-b+5}+\frac{1}{c^2-c+5}+\frac{1}{d^2-d+5}\leq\frac{4}{5}.$$

Problem 2 ⋆⋆ Non-negative reals a, b, c satisfy $a+b+c=3$ and $ab+bc+ca=2$. Find the minimum and the maximum of $a^4+b^4+c^4$.

Problem 3 ⋆⋆⋆ Prove that for all positive reals a, b, c,

$$\frac{a^2}{b+c}+\frac{b^2}{c+a}+\frac{c^2}{a+b}\geq\frac{a+b+c}{2}+\frac{(a-b)^2+(b-c)^2+(c-a)^2}{a+b+c}.$$

Problem 4 ⋆⋆⋆ Prove that for all positive reals a, b, c,

$$\frac{a(a^2+bc)}{b+c}+\frac{b(b^2+ca)}{c+a}+\frac{c(c^2+ab)}{a+b}\geq a^2+b^2+c^2.$$

Problem 5 ⋆⋆⋆⋆ Prove that for all real numbers a, b, c,

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$$(a+b+c)^4 + 16(ab+bc+ca)^2 \geq 7(a+b+c)^2(ab+bc+ca) + 12abc(a+b+c),$$

and determine all cases of equality.

Problem 6 ⋆⋆⋆⋆ Find the largest real number k such that

$$a+b+c+k\,abc \geq 3+k$$

for all non-negative reals a, b, c with $ab+bc+ca = 3$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 1994/95 Problem 3** ⋆⋆
   The three numbers are the roots of a cubic with two known coefficients. Sketch its graph.
2. **BMO2 2003/04 Problem 3** ⋆⋆
   With the sum equal to zero, express the power sums through the elementary symmetric functions.
3. **BMO2 2007/08 Problem 1** ⋆⋆
   Write both the condition and the target in terms of $x+y+z$ and $xy+yz+zx$.
4. **IMO 1984 Problem 1** ⋆⋆⋆
   Fix the sum and $xy+yz+zx$. The expression is linear in xyz.
5. **USAMO 2001 Problem 3** ⋆⋆⋆
   Write the condition and the target in p, q, r. The condition is linear in abc, so decide what to fix.
6. **IMO 1983 Problem 6** ⋆⋆⋆
   Ravi substitution, then look for an SOS form with positive coefficients.
7. **IMO 2001 Problem 2** ⋆⋆⋆⋆
   Look for an exponent t making each term at least $a^t/(a^t+b^t+c^t)$, which gives a tangent line bound for a sum that does not separate.
8. **USAMO 2017 Problem 6** ⋆⋆⋆⋆
   Bound each fraction below by a polynomial in the variable in the denominator. The minimum is not at the centre.

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4 Hints

1. Find the tangent line at $x = 1$ and the set of x on which the line lies above the curve. A variable outside that set makes its own term very small.
2. Express $a^4 + b^4 + c^4$ in terms of abc, then find the range of abc from the graph of $x^3 - 3x^2 + 2x$.
3. Write $\frac{a^2}{b+c} = \frac{a(a+b+c)}{b+c} - a$ and use the SOS form of $\sum \frac{a}{b+c} - \frac{3}{2}$. One coefficient may be negative.
4. Subtract a^2 from the first term, b^2 from the second and c^2 from the third, and factorise.
5. In terms of p, q, r the difference is linear in r, and the variables are real, so both ends of the range of r come from triples with two equal entries.
6. Test triples with one variable zero to find the candidate. For the proof, fix $a + b + c$. The left side is increasing in abc, so the lower end of the range of abc is what matters.

5 Solutions

Problem 1

Let a, b, c, d be real numbers (not necessarily positive) with $a + b + c + d = 4$. Prove that

$$\frac{1}{a^2 - a + 5} + \frac{1}{b^2 - b + 5} + \frac{1}{c^2 - c + 5} + \frac{1}{d^2 - d + 5} \leq \frac{4}{5}.$$

Let $f(x) = \frac{1}{x^2 - x + 5}$. Since $x^2 - x + 5 = (x - \frac{1}{2})^2 + \frac{19}{4} \geq \frac{19}{4}$, f is defined everywhere and $0 < f(x) \leq \frac{4}{19}$. We have $f(1) = \frac{1}{5}$ and $f'(1) = -\frac{1}{25}$, so the candidate line is $y = \frac{6-x}{25}$, and

$$\frac{6 - x}{25} - f(x) = \frac{(6 - x)(x^2 - x + 5) - 25}{25(x^2 - x + 5)} = \frac{-x^3 + 7x^2 - 11x + 5}{25(x^2 - x + 5)} = \frac{(x - 1)^2(5 - x)}{25(x^2 - x + 5)}.$$

Hence $f(x) \leq \frac{6-x}{25}$ for every real $x \leq 5$.

If $a, b, c, d \leq 5$, adding the four bounds gives a total of at most $\frac{24-(a+b+c+d)}{25} = \frac{4}{5}$. Otherwise some variable, say a, exceeds 5. Then $a^2 - a + 5 = a(a - 1) + 5 > 25$, so $f(a) < \frac{1}{25}$, and each of the other three terms is at most $\frac{4}{19}$. The total is less than $\frac{1}{25} + \frac{12}{19} = \frac{319}{475} < \frac{380}{475} = \frac{4}{5}$.

Equality requires the first case with $(x - 1)^2(5 - x) = 0$ for each variable, so each variable is 1 or 5. As they add up to 4, all four are 1. So equality holds only at $a = b = c = d = 1$.

Problem 2

Non-negative reals a, b, c satisfy $a + b + c = 3$ and $ab + bc + ca = 2$. Find the minimum and the maximum of $a^4 + b^4 + c^4$.

Let $r = abc$. Then $a^2 + b^2 + c^2 = 9 - 4 = 5$ and $a^2b^2 + b^2c^2 + c^2a^2 = (ab + bc + ca)^2 - 2abc(a + b + c) = 4 - 6r$, so

$$a^4 + b^4 + c^4 = 5^2 - 2(4 - 6r) = 17 + 12r.$$

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By Theorem 5, r ranges over the values for which $h(x) = r$ has three real roots, where $h(x) = x^3 - 3x^2 + 2x = x(x-1)(x-2)$, intersected with $r \geq 0$. Here $h'(x) = 3x^2 - 6x + 2$ vanishes at $x = 1 \pm \frac{1}{\sqrt{3}}$. With $u = \frac{1}{\sqrt{3}}$, $h(1-u) = (1-u)(-u)(-1-u) = u(1-u^2) = \frac{2\sqrt{3}}{9}$, and similarly $h(1+u) = -\frac{2\sqrt{3}}{9} < 0$. So $0 \leq r \leq \frac{2\sqrt{3}}{9}$.

The value $r = 0$ comes from the roots of h, that is $(a, b, c) = (0, 1, 2)$, which gives $a^4 + b^4 + c^4 = 17$. The value $r = \frac{2\sqrt{3}}{9}$ comes from the double root $1 - \frac{1}{\sqrt{3}}$ and the third root $3 - 2\left(1 - \frac{1}{\sqrt{3}}\right) = 1 + \frac{2}{\sqrt{3}}$, all non-negative, and gives $17 + 12 \cdot \frac{2\sqrt{3}}{9} = 17 + \frac{8\sqrt{3}}{3}$. Since $a^4 + b^4 + c^4 = 17 + 12r$ is increasing in r, the minimum is 17 and the maximum is $17 + \frac{8\sqrt{3}}{3}$.

Answer: Minimum 17, maximum $17 + \frac{8\sqrt{3}}{3}$

Problem 3

Prove that for all positive reals a, b, c,

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2} + \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{a+b+c}.$$

Let $p = a + b + c$. Since $\frac{a^2}{b+c} = \frac{ap}{b+c} - a$, the left side equals $p \sum \frac{a}{b+c} - p$, so

$$\sum \frac{a^2}{b+c} - \frac{p}{2} = p\left(\sum \frac{a}{b+c} - \frac{3}{2}\right) = \sum \frac{p\,(a-b)^2}{2(a+c)(b+c)}.$$

Subtracting $\sum \frac{(a-b)^2}{p}$, the inequality becomes $S = \sum S_c (a-b)^2 \geq 0$ with

$$S_c = \frac{p}{2(a+c)(b+c)} - \frac{1}{p} = \frac{p^2 - 2(a+c)(b+c)}{2p(a+c)(b+c)} = \frac{a^2 + b^2 - c^2}{2p(a+c)(b+c)},$$

because $p^2 - 2(ab + bc + ca) - 2c^2 = a^2 + b^2 - c^2$. The expression is symmetric, so we may assume $a \geq b \geq c$. Then $S_c \geq 0$ and $S_b = \frac{a^2 + c^2 - b^2}{2p(a+b)(b+c)} \geq 0$, but S_a may be negative (for instance when a is large). We use criterion (2):

$$2p\,(S_a + S_b) = \frac{(b^2 + c^2 - a^2)(b+c) + (a^2 + c^2 - b^2)(a+c)}{(a+b)(b+c)(c+a)} = \frac{(a+b)(a-b)^2 + c^2(a+b) + 2c^3}{(a+b)(b+c)(c+a)} \geq 0,$$

where the numerator was expanded as $a^3 + b^3 - a^2 b - ab^2 + c^2(a+b) + 2c^3$ and $a^3 + b^3 - a^2 b - ab^2 = (a+b)(a-b)^2$. Also $S_b + S_c \geq 0$. By criterion (2), $S \geq 0$.

For equality. The proof of criterion (2) gives $S \geq (S_a + S_b)(b-c)^2 + (S_b + S_c)(a-b)^2$, and for positive reals $S_a + S_b > 0$ (because of the term $2c^3$) and $S_b + S_c \geq S_c > 0$. So equality holds only at $a = b = c$. The constant in front of the last fraction is best possible. With $a = b = 1$ and $c \to 0$ both sides tend to 2. This doubles the error term in Practice 6 of L4-AL2, where all three coefficients were non-negative and criterion (1) sufficed.

Problem 4

Prove that for all positive reals a, b, c,

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$$\frac{a(a^2+bc)}{b+c}+\frac{b(b^2+ca)}{c+a}+\frac{c(c^2+ab)}{a+b}\geq a^2+b^2+c^2.$$

We have

$$\frac{a(a^2+bc)}{b+c}-a^2=\frac{a\left(a^2+bc-ab-ac\right)}{b+c}=\frac{a(a-b)(a-c)}{b+c},$$

and similarly for the other two terms. So the inequality is

$$x(a-b)(a-c)+y(b-a)(b-c)+z(c-a)(c-b)\geq 0,\qquad x=\frac{a}{b+c},\ y=\frac{b}{c+a},\ z=\frac{c}{a+b}.$$

The expression is symmetric, so we may assume $a\geq b\geq c$. Then $x\geq y$, because $a(c+a)-b(b+c)=(a-b)(a+b)+c(a-b)\geq 0$. All of x,y,z are positive, so Theorem 4 (Vornicu-Schur) gives the inequality.

For equality, the proof of Theorem 4 with $x\geq y$ gives a left side of at least $y(a-b)^2+z(a-c)(b-c)$, with $y,z>0$. So equality forces $a=b$ and $(a-c)(b-c)=0$, that is $a=b=c$, and $a=b=c$ gives equality. (As with Schur's inequality, the sum also tends to 0 at $(1,1,0)$, which is not allowed here.)

Problem 5

Prove that for all real numbers a,b,c,

$$(a+b+c)^4+16(ab+bc+ca)^2\geq 7(a+b+c)^2(ab+bc+ca)+12abc(a+b+c),$$

and determine all cases of equality.

Let $F=p^4-7p^2q+16q^2-12pr$. We must show $F\geq 0$. For real numbers $p^2-3q=\frac{1}{2}\sum(a-b)^2\geq 0$.

First we evaluate F on a triple (t,t,s). Then $p=2t+s$, $q=t^2+2ts$, $r=t^2s$, and a direct expansion gives

$$F(t,t,s)=(s-t)^2(s-2t)^2.$$

(As a check, $s=0$ gives $16t^4-28t^4+16t^4=4t^4$ on both sides, and $s=t$ gives 0.) In particular $F\geq 0$ on every triple with two equal entries, and $F=0$ there exactly when the third entry equals the other two or is twice them.

Now take any real triple. If $p^2=3q$, then $a=b=c$, which is a triple of the form (t,t,t), so $F=0$. Suppose $p^2>3q$, and fix p and q. By Theorem 5(1) the possible values of r form an interval $[r_1,r_2]$, and the triples with $r=r_1$ or $r=r_2$ have two equal entries, so $F\geq 0$ at both ends. As a function of r, F is linear with slope $-12p$. If $p=0$, then $F=16q^2$, and $q<0$ because $p^2>3q$, so $F>0$. If $p\neq 0$, F is a non-constant linear function of r on $[r_1,r_2]$, so F is at least its value at one end, which is non-negative, and F can equal 0 only if r is that end and F vanishes there.

Hence $F\geq 0$ for all real a,b,c, and in every case of equality the triple has two equal entries t and a third entry s with $(s-t)^2(s-2t)^2=0$. So equality holds exactly when $a=b=c$, or when (a,b,c) is a permutation of $(t,t,2t)$ for some real t, such as $(1,1,2)$. Note that no AM-GM argument can prove this. The variables may be negative and the equality case $(1,1,2)$ is not at the centre. The sign of p decides which end of the interval is the minimum, but since both ends are checked, we never needed to know which.

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Problem 6

Find the largest real number k such that

$$a+b+c+k\,abc \geq 3+k$$

for all non-negative reals a, b, c with $ab+bc+ca=3$.

The answer is $k = 2\sqrt{3}-3$. The triple $(\sqrt{3}, \sqrt{3}, 0)$ satisfies the condition and gives $2\sqrt{3} \geq 3+k$, so $k \leq 2\sqrt{3}-3$.

Now let $k = 2\sqrt{3}-3 > 0$, write $p = a+b+c$, $r = abc$, and $F = p+kr-3-k$. Fix p. The condition fixes $q = 3$. If $p^2 = 3q$ then $a = b = c = 1$ and $F = 0$. Otherwise F is increasing in r, so by Theorem 5(2) we only need to prove $F \geq 0$ when one variable is 0 or two variables are equal.

If $c = 0$, then $ab = 3$ and $p = a+b \geq 2\sqrt{ab} = 2\sqrt{3}$, so $F \geq 2\sqrt{3}-3-k = 0$.

If $(a, b, c) = (t, t, s)$, the condition $t^2+2ts = 3$ gives $t > 0$ and $s = \frac{3-t^2}{2t}$, and $s \geq 0$ gives $t \leq \sqrt{3}$. Then $r = t^2 s = \frac{t(3-t^2)}{2}$ and

$$2t\,F = 4t^2 + (3-t^2) + kt^2(3-t^2) - 2t(3+k) = 3(t-1)^2 - kt(t^3-3t+2) = (t-1)^2\big(3-kt(t+2)\big),$$

using $t^3-3t+2 = (t-1)^2(t+2)$. For $0 < t \leq \sqrt{3}$ we have $t(t+2) \leq \sqrt{3}(\sqrt{3}+2) = 3+2\sqrt{3}$, and $k(3+2\sqrt{3}) = (2\sqrt{3}-3)(2\sqrt{3}+3) = 3$. So $3-kt(t+2) \geq 0$ and $F \geq 0$.

Hence the inequality holds for $k = 2\sqrt{3}-3$, with equality at $(1, 1, 1)$ and at the permutations of $(\sqrt{3}, \sqrt{3}, 0)$, and fails for every larger k. (For smaller k the inequality also holds, since $abc \leq 1$ under the condition, so the inequality only gets weaker as k decreases.)

Answer: $k = 2\sqrt{3}-3$